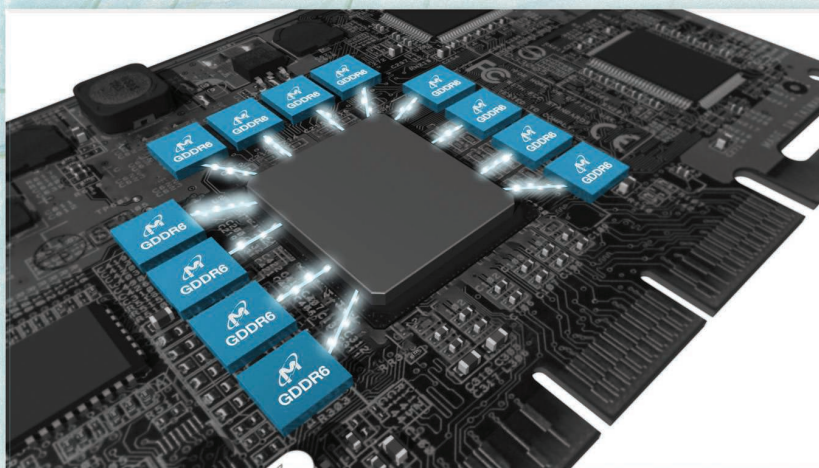


Exhibit CC-108

Exhibit 10

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Recent Evolution in the DRAM Interface

Mile-markers along memory lane

As semiconductor technologies have matured and computing applications have proliferated, memory technology has experienced a corresponding evolution and diversification to address application-specific requirements and topological targets. These include the following, not necessarily in chronological order.

- NAND and NOR flash have addressed nonvolatile memory needs, paving

the way for exciting applications like handheld devices and solid-state drives (SSDs).

- Dynamic random-access memory (DRAM) has also evolved to satisfy a variety of system-level emphases.
- Three-dimensional crosspoint technology, just now going public, has stepped forward to fill the feature and function space between NAND and DRAM.
- Alternative emerging memory mechanisms are continuously evaluated with hopes either to combine the best features of existing technologies or simply to identify a path with viable, prolonged scalability.

While shrinking memory cell size has fueled the memory industry for decades, accessibility to that ever-expanding memory capacity has also grown through distinct architectural embodiments to optimize latency, capacity, and/or power, along with an ever-increasing demand for external bandwidth.

Interestingly, with very few exceptions, system memory access has leaned on single-ended signaling, with DRAM generally leading the way in signaling advancements, including addressing the metrics most interesting to the input/output (I/O) circuit designer and signal integrity (SI) engineer (e.g.,

increasing frequency and downward voltage scaling). At the same time, interfaces to NAND and NOR flash memories have lagged behind, hampered in part by capacity demands, resulting in intrinsically slow communication to many parallel memory loads.

This article provides an introduction to the development of the DRAM interface, beginning with a brief historical account, highlighting key milestones in the evolution of both circuit and interconnect, leading up to the presently ramping generation of the three most familiar DRAM embodiments or standards, namely, fifth-generation double data rate (DDR5), low-power DDR5 (LPDDR5), and sixth-generation graphics DDR (GDDR6), while acknowledging the technical value of alternative DRAM solutions, past and present. Although the topic of high-speed signaling often suggests fully differential, serializer/deserializer (SerDes) architecture and the memory interface design may appear mundane, the driving factors and performance-impeding constraints of memory applications have fostered a rich history.

While each DDR, LPDDR, and GDDR generation introduced new features and architectural advancements, including macrolevel interface enhancements (optimized for target markets) such as dual-channel support or changes in memory control protocols to facilitate higher command bandwidth over a reduced number of pins, the focus of this article is on the scaling of bandwidth/pin, based on enhancements to the pin electronics in conjunction with the physical/electrical constraints of the off-chip interconnect. The intent is not to oversimplify significant advances in DDR technology but rather to highlight those most recent signaling-related developments across the three primary embodiments.

Background

In the year 2000, the Joint Electron Device Engineering Council (JEDEC) released the first DDR synchronous DRAM standard (DDR-200), boasting a per-pin data rate of 200 Mb/s. Yet,

In 2019, the computing world is in the process of ramping the newly introduced GDDR6 graphics DRAM, with I/O bandwidth reaching as high as 16 Gb/s/pin.

even before the release of this foundational DDR standard, platforms consuming DRAM were already diverse enough to warrant a degree of application-specific optimization beyond the baseline. The graphics industry, in particular, was already clamoring for greater bandwidth, and, as such, the release of the alternative GDDR (or synchronous graphics RAM) solution actually preceded the DDR release by a few months [1]. Soon after, the growing sensitivity to power consumption of mobile applications sparked the LPDDR offshoot, redesigned to function at lower supply voltages while also offering new low-power modes of operation (e.g., lower-power consumption when idle).

As shown in Figure 1, since their inception, this trio of architectures has continued along parallel paths of multigenerational evolution, while spawning an abundance of alternative flavors within the greater DDR family [2]. Scattered throughout the history of DDR, one also finds further

diversification through standards and architectures like reduced-latency DRAM, Rambus extreme data rate, Wide I/O (generations 1 and 2), the Hybrid Memory Cube (generations 1–3), and High Bandwidth Memory (HBM), which is presently in its second generation. Yet DDR, LPDDR, GDDR, and, more recently, HBM tend to dominate the present and near-term landscape.

As indicated, DRAM architectural adaptation has been accompanied by continuous per-pin data-rate scaling. Figure 2 compares the data-rate history of the DDR technologies, highlighting three distinct trajectories in the rate of speed scaling. (It is noted here that the width of the DDR parallel interface generally differs among the technologies and applications, and, thus, the per-pin perspective does not capture the typical effective system bandwidths supported by each.) While the memory industry has dabbled in differential signaling solutions over the years, the dominant DRAM interface protocols of

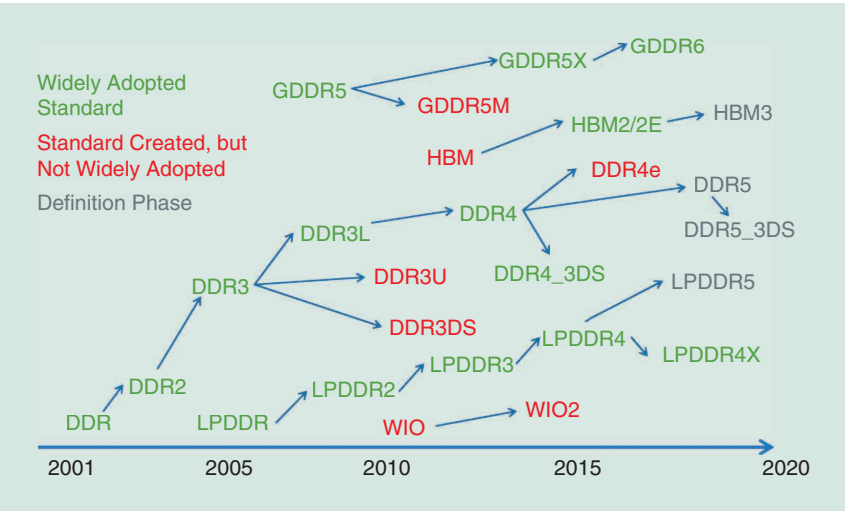


FIGURE 1: The diversification of DDR-based DRAM over the last two decades. GDDR 1–4 and non-JEDEC standards like Hybrid Memory Cube 1–3 are not shown. WIO: wide I/O.

One of the most compelling growth industries for GDDR is in automotive applications.

today, including those captured in Figure 1, remain single ended.

To better understand the disparity in per-pin bandwidth scaling, it is helpful to consider the high-volume applications typically associated with each technology. For example, GDDR is most often found in applications emphasizing bandwidth performance over capacity, power, and even cost of imple-

mentation (within reason). As a result, optimizing for high memory bandwidth includes, among other factors, shorter, cleaner channel design with point-to-point loading. Interestingly, mobile (LPDDR) applications ultimately converged on similar channel characteristics, not because they have explicitly targeted extreme speeds but because their sensitivity to footprint (e.g., the

miniaturization of applications) leads to shorter, simpler interconnects.

While multiple mobile topologies exist, one of the most common mobile channels is the package-on-package (POP) configuration, in which the solder balls of the memory package land directly on the top of the processor package substrate. In contrast to GDDR and LPDDR, DDR has been the workhorse of the family and, as such, has functioned over the greatest range of operating conditions, including longer, more complicated, and more heavily loaded channels. These distinctions in operating environments, coupled with market demands, have clearly influenced the per-pin bandwidth deltas presented in Figure 2.

Figure 3 compares the insertion loss characteristics for an exemplary multidrop DDR application, a POP LPDDR interface, a typical graphics card GDDR environment, and the silicon interposer-based interconnect associated with HBM. Consistent with the preceding discussion, both the GDDR and LPDDR channels demonstrate fairly clean (low-loss) behavior out to about 10 GHz, while the multidrop channel, typical of a computer server application, rolls off more quickly and introduces a large null in the channel response at low frequencies. In contrast to the DDR channels, the large low-frequency offset in the HBM channel is characteristic of highly resistive silicon interposer transmission lines corresponding to the uniquely aggressive HBM trace width, space, and thickness (e.g., $2\mu\text{m} \times 2\mu\text{m} \times 2\mu\text{m}$).

To better understand why LPDDR has not tracked GDDR in speed scaling, despite the comparable channel characteristics, it is noted that the LPDDR channel behavior reported in Figure 3 does not account for the steeper roll-off experienced under elevated loading conditions, when multiple DRAMs are connected in parallel within the package to increase the overall system memory capacity. It is also important to remember that a driving principle in mobile applications is to maximize

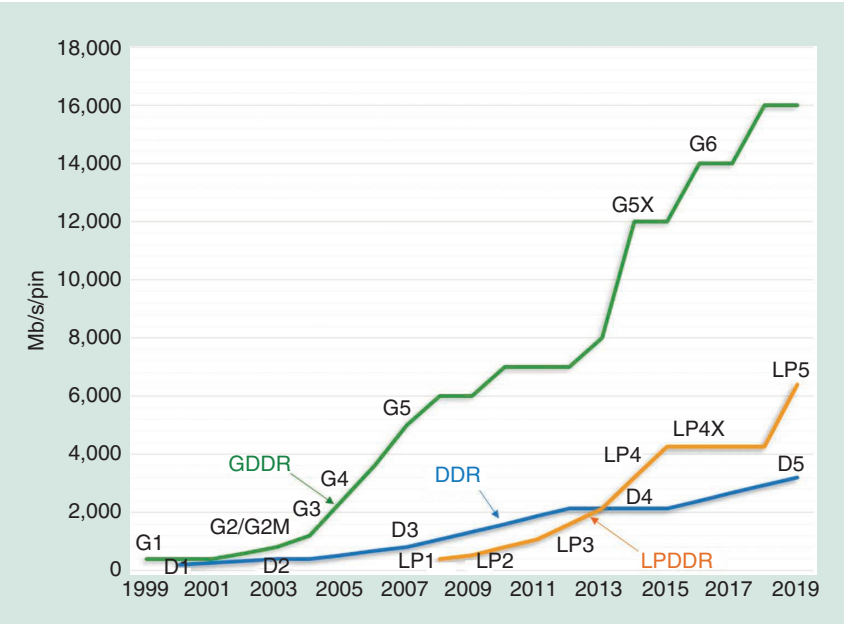


FIGURE 2: The DRAM per-pin bandwidth history for the DDR family.

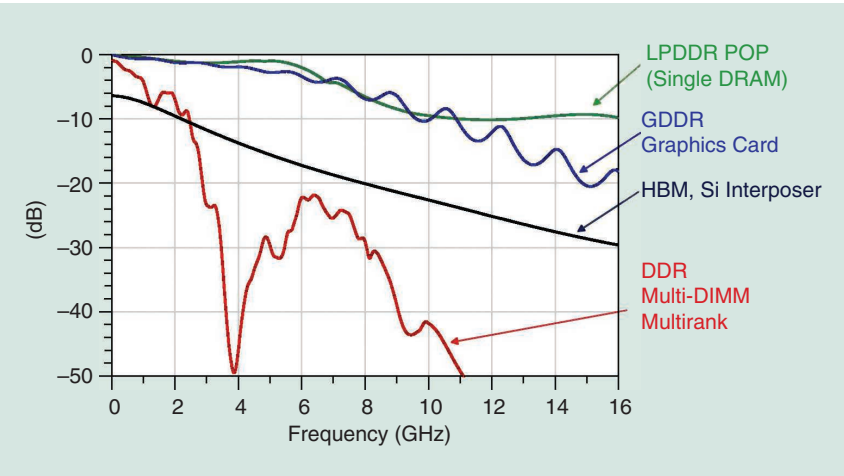


FIGURE 3: A comparison of the distinct channel characteristics of common DDR-based DRAM applications.

performance while limiting battery power consumption (and cost). The motivation to lower memory subsystem power in mobile has led to a more rapid reduction in voltage scaling, while still maintaining a healthy signaling margin within DDR, clarifies that LPDDR is, in fact, nearly aligned with GDDR in terms of composite signal margin, which is a function of both decreasing voltage and decreasing cycle or bit unit interval (UI). While the GDDR6 UI has narrowed to 62.5 ps, it continues to operate from a 1.35-V voltage supply. In contrast, the LPDDR5 UI will soon reach 156.25 ps, but the output supply voltage (VDDQ) will simultaneously drop to 0.5 V.

Based on the history in Figure 4, one might consider that DDR5 speed scaling is unnecessarily lagging behind. However, the levels reported in Figure 4 do not comprehend the margin impact of the channel responses shown in Figure 3. When the complexity of the typical multidrop DDR interconnect is accounted for, the signaling margins of DDR5 drop off the bottom of the graph. In fact, it is anticipated that the data eye presented to the DDR5 DRAM near end-of-life speeds will be completely closed. In all, considering the aggressive loading, voltage, and speed constraints imposed on these technologies, the state of DDR, LPDDR, and GDDR signaling in 2019 is impressive.

DDR in 2019

As indicated in the previous figures, in 2019, the computing world is in the process of ramping the newly introduced GDDR6 graphics DRAM, with I/O bandwidth reaching as high as 16 Gb/s/pin [3], as well as the just-introduced LPDDR5 [4], with the first hints of DDR5 finding their way into the public domain [5]. Each of the current standards incorporates new features and functions to address present and foreseeable market needs—including new markets, such as autonomous vehicles and other applications driven by

advances in artificial intelligence (AI) and machine learning (ML)—as well as an accumulation of innovation built up over the preceding generations.

From a circuit design perspective, the historical scaling of DDR bandwidth has not been driven by major advances in transistor processing at each technology node, but rather has been enabled through the adoption of many obstacle-mitigating methods, despite limited transistor performance scaling. One exemplary enhancement, among many adopted by the fourth generation of all three DDR segments, was data bus inversion (DBI) [6], [7]. It was introduced to address the simultaneous switching output (SSO) noise [8] associated with several single-ended drivers transitioning together in the same direction, leading to large fluctuations in the current demanded from the local power supply nodes. Technically, DBI was integrated into certain GDDR products by the second generation, yet was not available universally until the fourth [9]. DBI provides a relatively simple and low-overhead approach to cut the instantaneous current in half while also lowering the average I/O current, although to

a lesser degree. Thus, DBI has been a key enabler of interface scaling (see Figure 5), while adding the fringe benefit of power reduction.

A second recent change, intended to overcome transistor bandwidth limitations in the DRAM input path, has been the transformation of the DRAM input architecture from strictly source synchronous to a meso-synchronous solution. GDDR5, LPDDR4, and now DDR5 data rates have passed the point at which full-speed on-die buffering of wideband data to fully match the end-to-end data and clock paths has become inefficient, if not impossible. However, because of the narrow-band nature of the forwarded clock, it is still possible to distribute the data-latching clock to each data pad and capture the data immediately as they enter the DRAM. Loss of high-speed noise and jitter correlation, resulting from the unmatched paths, is much less a problem than the data-dependent jitter incurred along the on-die, bandwidth-limited data path of the earlier source-synchronous architecture. Of course, such a change has introduced an additional layer of interface calibration complexity to center the sample time

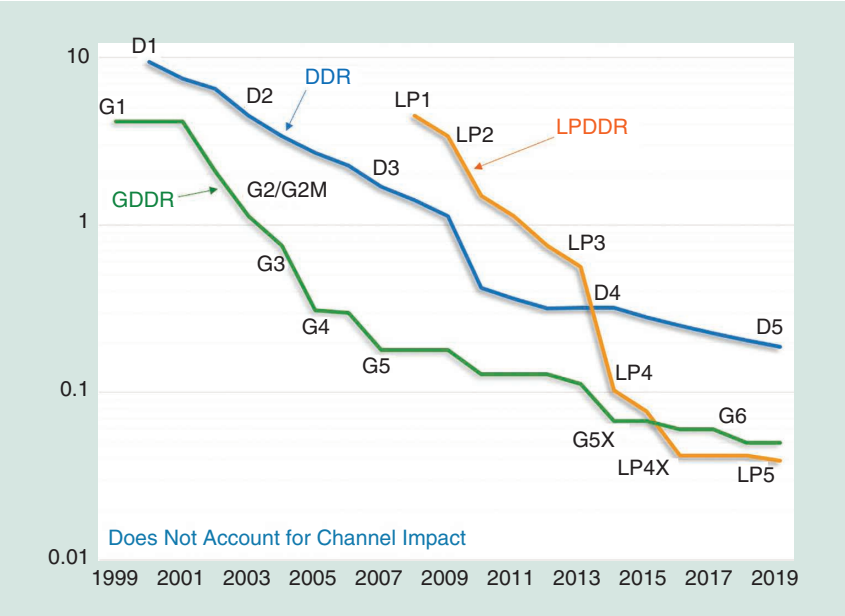


FIGURE 4: The DDR-based DRAM nominal signaling margin (transmit swing in volts x unit interval in picoseconds.) due to increasing frequency and decreasing output swing (correlated with the I/O supply voltage).

within the shrinking data eye, a necessary tradeoff.

Before diving deeper into a few of the more unique nuances of each of the DDR technologies, it is interesting to highlight that, despite a significant divergence in many respects, an enabling commonality has recently emerged, namely, that for the first time a common memory capacity (8 Gb) is currently available in DDR, LPDDR, and GDDR. As a result, today's system architect has options to boost bandwidth (GDDR) or lower

power (LPDDR) while still efficiently providing the target system memory capacity. This is not meant to suggest that there is no longer a market for traditional DDR. Indeed, new features available in DDR5 enable its use over a much broader physical/topological design space, as discussed in the following section.

DDR5

As indicated in the introductory section, DDR5 might well be called the workhorse of DRAM, as it will

serve the widest range of operating and environmental conditions, from powering data-center servers that enable the cloud to powering smart appliances in the kitchen. From an interconnect perspective, DDR5 may find itself signaling across a short and benign channel in a variety of applications, including data buffering in an SSD, but is more likely to find itself mounted to a dual inline memory module (DIMM) inserted into one of multiple connectors on a high-capacity server motherboard. The most common system configuration is the two DIMM-per-channel, dual-rank (two loads per DIMM), in which each of two DIMMs is configured to connect two DRAM loads per wire, with both DIMMs connected to the same controller channel, for a total of four loads.

In this configuration, only one DRAM rank at a time may utilize the memory bus, as directed by the command and control pins. When one DRAM is actively transmitting or receiving over the bus, the other three ranks introduce signal reflections through poor termination of the channel at their corresponding locations. Based on the distance between them, reflected energy accumulates and tends to degrade the signal at the active I/O. While the exact impact of the reflections depends on the relationship between the signal propagation time and data frequency, a somewhat systematic trend in relative eye quality is often observed. As depicted in Figure 6, it is possible for the rank furthest from the controller (the host) to experience four times the signaling margin when compared with the nearest rank.

As a result of such environmental factors, many of the DDR adaptations have been more macrolevel, such as modifications to the DIMM itself, as highlighted in Figure 7. In the unbuffered DIMM design, data, command/address/control, and clock signals flow directly between the controller and the DRAM [10]. In a later effort to shorten and simplify the overall command/address/

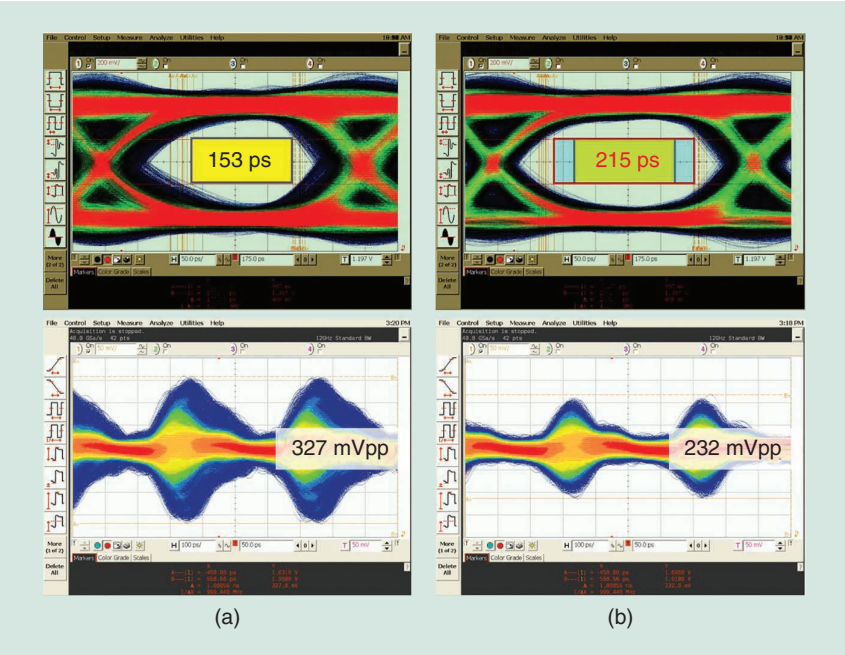


FIGURE 5: The measured impact of DBI on GDDR4 signaling: (a) unencoded and (b) DBI encoded. The 2.5-Gb/s data eyes are at the top, and the corresponding I/O power supply noise is below.

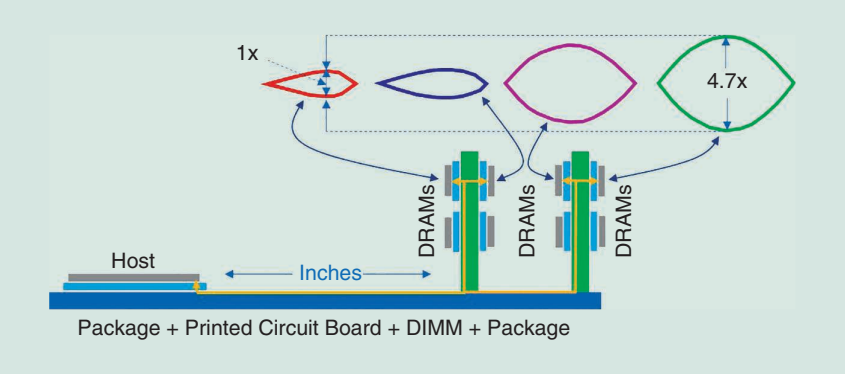


FIGURE 6: Examples of worst-case write data eye contours (3.2 Gb/s, no crosstalk, no equalization) over a typical DDR4 multidrop server channel, highlighting the systematic difference in eye margins at each DRAM rank, primarily attributed to location-specific signal reflections and crosstalk.

control and clock signal paths, a register was introduced on the registered DIMM to intercept and retransmit those signals out to the DRAMs, resulting in a substantial signal margin increase at each DRAM location [11]. A third topology introduced a buffer into the data path to effectively eliminate the effects of reflections and channel loss, as far as the DRAM is concerned. In this configuration, also known as the *load-reduced DIMM*, the DRAM communicates over a significantly simplified channel between the data buffer and the memory [12]. Both the register and the data buffers are implemented in logic processes with more capable transistors. Thus, while

these architectural changes do not eliminate the signal-degrading physics of the multidrop channel, the more challenging signaling is addressed by the more capable circuits on the register, data buffers, and host controller or processor, rather than the DRAM.

In addition to macrolevel system accommodations held over from DDR3 and DDR4, the DDR5 die architecture introduces several circuit-based features to address the challenging signaling environment. Perhaps the most profound enhancement is the integration of four taps of decision feedback equalization (DFE) at the DRAM data path input [5]. While several equalization methods might

have been adopted, DFE was chosen based on the nature and severity of the multidrop interconnect previously referenced.

Figure 8 helps clarify why DFE was chosen over the more power-efficient and less complicated de-emphasis and/or continuous-time linear equalization (CTLE) alternatives. In Figure 8(a) and (b), the red curve represents the unequalized channel response. Figure 8(a) demonstrates a smooth roll-off indicative of a well-designed but lossy channel, while Figure 8(b) corresponds to the multidrop system topology discussed with reference to Figure 6. In both cases, the cyan curve captures the composite channel/CTLE

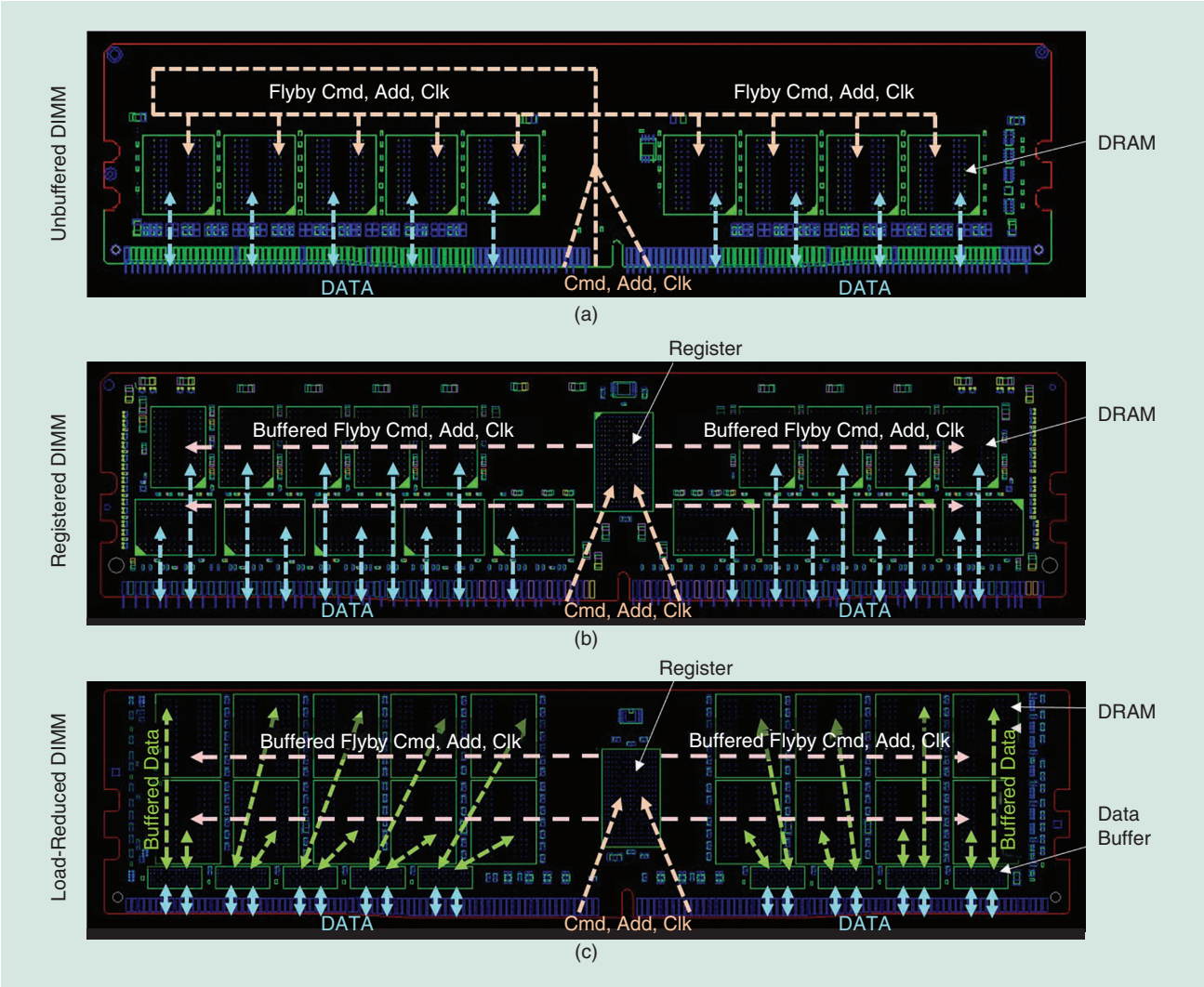


FIGURE 7: Floorplans capturing the basic topologies of the (a) unbuffered DIMM, (b) registered DIMM, and (c) load-reduced DIMM, while highlighting key signal paths. DIMM fingers (pins) at the bottom of each image connect into the main motherboard routing. CLK: clock; Cmd: command; Add: address.

response, while the green curve corresponds to the composite channel/de-emphasis response, and the blue curve presents the composite channel/DFE response. As shown in Figure 8(a), all three equalizers are capable of flattening the composite response to a frequency beyond 1 GHz (further extension is possible). In this case, any of the solutions might suffice, with the final design choice depending on other factors, such as the circuit power or silicon footprint. On the other hand, Figure 8(b) demonstrates that neither CTLE nor de-emphasis is capable of compensating for the insertion loss null resulting from the excessive signal reflections in the multidrop memory configuration.

The DFE, however, is able to span the null, providing a significant increase in received SI.

Figure 9 provides the more familiar time-domain presentation of the unequalized channel (red) and composite channel/equalizer responses (cyan = CTLE, green = de-emphasis, and blue = four-tap DFE) for the multidrop server channel response reported in Figure 8(a) and represented in Figure 6. As shown, both CTLE and de-emphasis suppress the intersymbol interference (ISI), as measured by the reduced variation of the high and low signal levels (e.g., less voltage distribution) relative to the unequalized channel, yet neither opens the eye. In contrast, the

four-tap DFE demonstrates a clear ability to open the data eye, despite the complex channel behavior.

The tap-count choice, while ultimately imposed by the standard, was specified intelligently as a function of both the anticipated channel behaviors and circuit design tradeoffs (including area, power, and routing congestion to each DFE block). Figure 10 presents the channel-based argument, while Figure 11 highlights a portion of the circuit considerations. Figure 10 compares the target response of an ideal zero-forcing equalizer (corresponding to the multidrop channel previously introduced), with the simulated DFE response for one tap, four taps, and 10 taps. As the DFE tap count

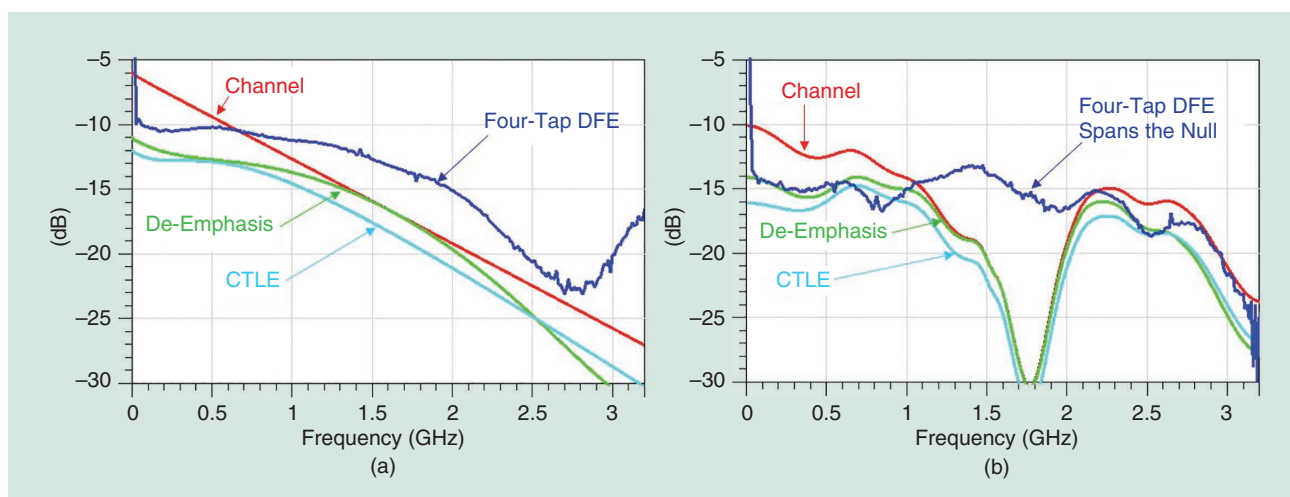


FIGURE 8: The effectiveness of CTLE, de-emphasis, and four-tap DFE applied to (a) a well-behaved low-pass channel and (b) a typical multidrop server channel.

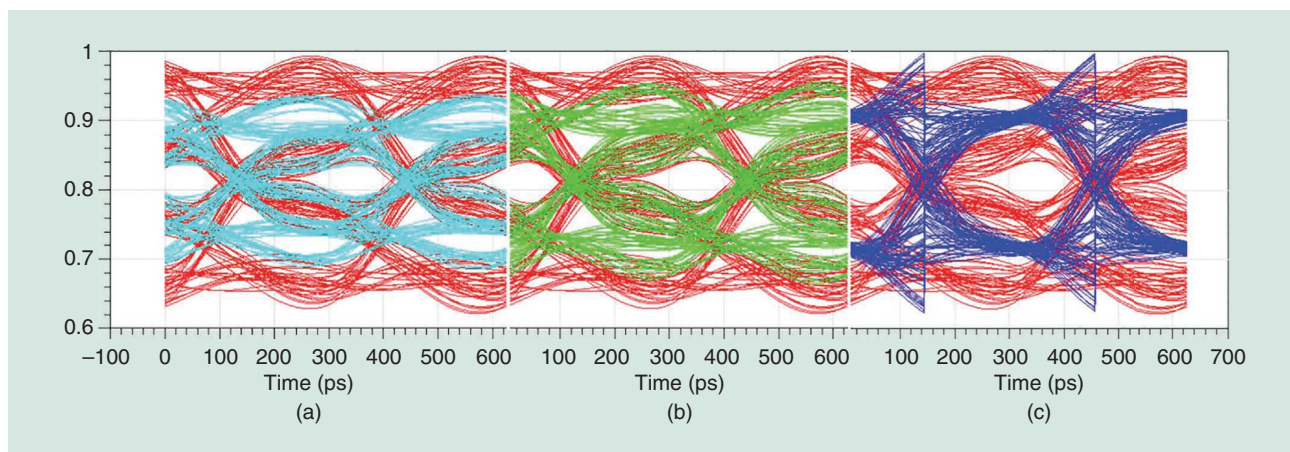


FIGURE 9: A time-domain comparison of the relative equalizer effectiveness over the multidrop server channel presented in Figure 8(b): (a) CTLE, (b) de-emphasis, and (c) four-tap DFE.

increases, the simulated DFE response approaches the ideal behavior. Clearly, the four-tap response more closely approximates the target behavior over the lower frequencies than the single-tap configuration, while the 10-tap design is hardly distinguishable from the four-tap case. A similar analysis could be carried out to the same end in the time domain through pulse response evaluation.

Having conceded that a one-tap solution would leave substantial margin on the table and that even two taps might not span the near-term ISI introduced by the multidrop channel, the next constraint is implementation. Figure 11 presents a block diagram for a four-tap, speculative DFE suitable for DDR5. As indicated, a pseudodifferential variable gain preamplifier compares the incoming data signal with a dc reference voltage V_{refDQ} , providing a fully differential signal to the four parallel paths of the DFE. Summers, preceding each of four regenerative

latches, use digital values tapped from the subsequent deserializer stage (Tap0, Tap180, Tap360, and Tap540) to modulate the signals from the pre-amplifier depending on the previous

data-capture decision, while each of the four latches is triggered by one of four data strobe phases (DQS0, DQS180, DQS360, and DQS540). Once a few bits have passed, all four taps

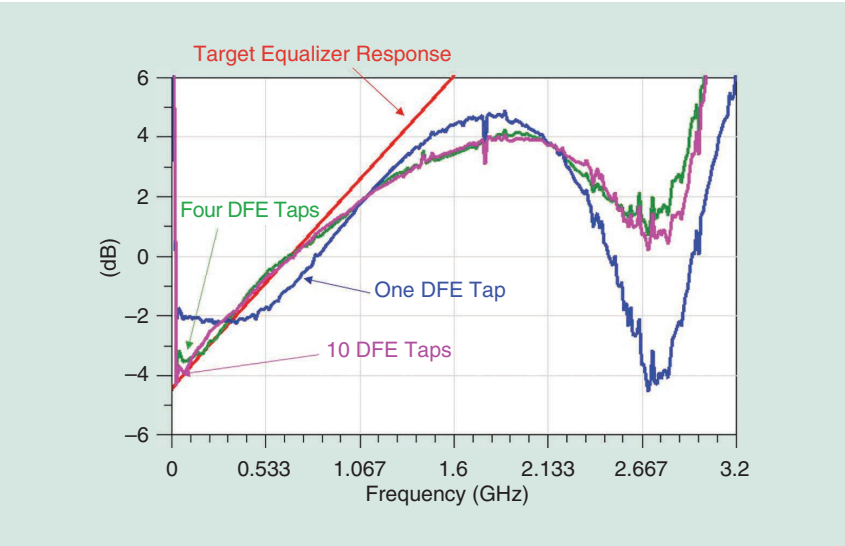


FIGURE 10: Simulated DFE responses for one-tap, four-tap, and 10-tap solutions and their relative approximations to the ideal zero-forcing equalizer.

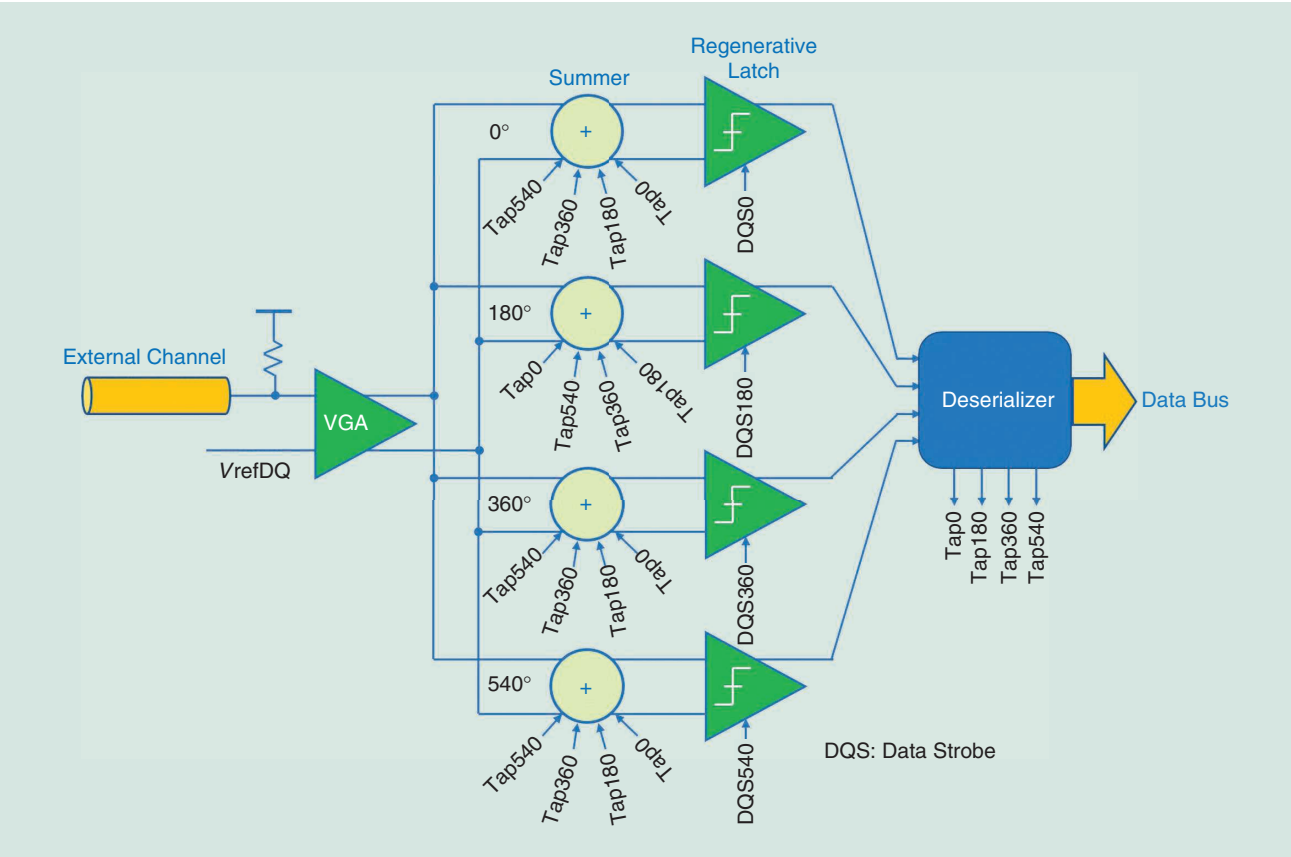


FIGURE 11: A basic functional schematic of a four-tap DFE suitable for DDR5. VGA: variable gain amplifier.

work in tandem to uniquely modulate the signal sequentially presented to each latch.

The basic structure of Figure 11 lends itself to scaling, but support for an odd number of taps is awkward and inefficient. The figure also makes clear that several timing-critical connections must be made between the various functional blocks at each data input. Given the preference for an even tap count, compatibility with a four-phase clocking solution (likely already designed into the interface), and the complexity and high wire-count of the four-tap embodiment, it was difficult to justify support for additional taps. At the same time, there was a concern that anything less robust might not provide sufficient compensation for the challenging DDR5 signaling conditions.

Figures 12 and 13 demonstrate the effectiveness of the four-tap DFE to open a nearly closed data eye. To facilitate this experiment, a golden channel test environment was designed

to produce significant ISI within the first four postcursors (e.g., the four-UI correctible range of the DFE). The green oscilloscope measurement [Figure 12(a)] presents the received data eye closure over several subsequent cycles, near the DRAM input, while the colored measurement in Figure 12(b) reports the same signal following four-tap DFE postprocessing within the oscilloscope.

The data eye shmoos in Figure 13 come from the DRAM itself, using loopback functionality to count errors during write-data capture. The eye shmoo in Figure 13(a) presents the pass/fail eye (green = pass, yellow/red = fail), as captured by the DRAM without the DFE enabled, while the measurement in Figure 13(b) presents the pass/fail eye with the DRAM's four-tap DFE enabled. While the correlation is not perfect, in part because the oscilloscope measurement point was located outside of the DRAM package (potentially degrading the SI), the match is close, with a 7-mV

oscilloscope measurement corresponding to a 10-mV passing eye on the DRAM and a 79-mV post-DFE oscilloscope measurement aligning well with the 90-mV post-DFE passing eye on the DRAM. A potential takeaway from this experiment is that the realized DRAM DFE functionality is comparable to the high-speed oscilloscope circuitry.

A secondary challenge born out of this multitap DFE integration is I/O modeling—in particular, how to share DDR5 I/O models with technology partners and customers. Traditionally, the sharing of DRAM I/O circuit models, while protecting intellectual property and simultaneously speeding up interface simulations, has been handled by the I/O Buffer Information Specification (IBIS), essentially a black box behavioral model based on circuit I - V characteristics. With the introduction of equalization into each of the DDR-based architectures, DRAM I/O modeling must expand in capability to support accurate system simulations. One approach to modeling the new equalization functionality is to adapt methods developed for the IBIS-Algorithmic Modeling Interface (IBIS-AMI), traditionally used to model equalization in high-speed SerDes [13], [14]. The IBIS-AMI incorporates the equalization functions into a compiled executable coupled with the traditional IBIS model. While this technique has worked well to simulate the eye opening and bit error ratio of equalized signals in clock data recovery (CDR)-enabled SerDes, there was no ready-made solution for modeling single-ended equalization in a meso-synchronous architecture.

In the typical IBIS-AMI simulation flow, the circuit impulse response is calculated through the derivative of its step response. For single-ended signals, the common-mode voltage of the signal is lost through this differentiation. Single-ended interfaces are also subject to SSO, and, as the IBIS-AMI simulation flow assumes a linear time-invariant (LTI) system, it does not lend itself to analyzing

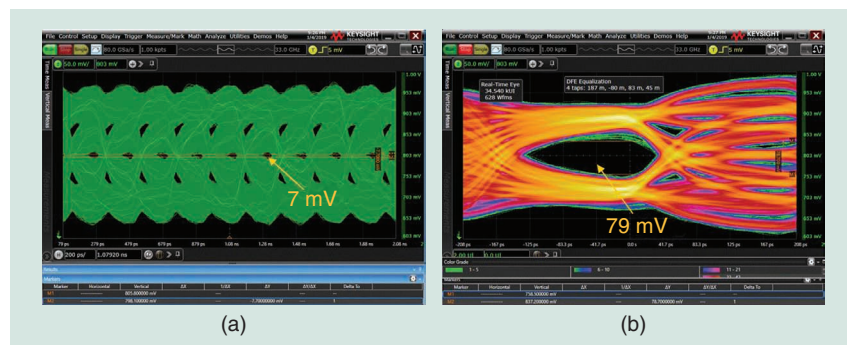


FIGURE 12: Oscilloscope measurements of stimulus and projected DFE performance. Input from a “golden channel” is used to stress (a) the DRAM DFE and (b) the “captured” data eye following four-tap DFE emulation within the oscilloscope.

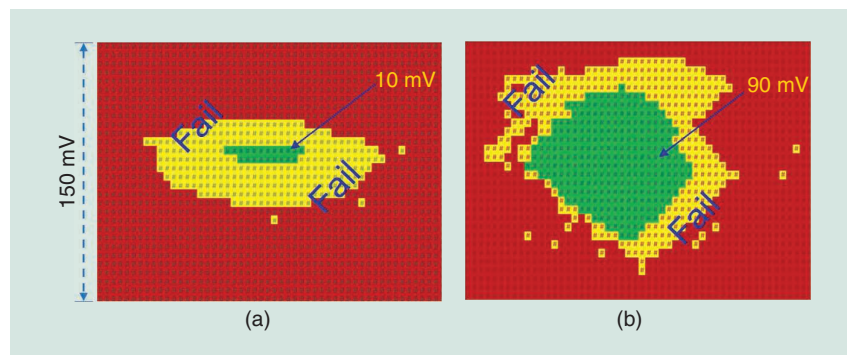


FIGURE 13: DRAM input data eye shmoos corresponding to the oscilloscope measurements in Figure 12: (a) without four-tap DFE enabled and (b) with four-tap DFE enabled.

power delivery effects, which may be thought of as time-varying impairments. In addition, DDR systems tend to be more nonlinear than traditional SerDes systems, as a result of transmitter output impedance variation over the operating range, mismatch between pull-up (PU) and pull-down (PD) impedances, and distinct rise and fall behaviors.

The IBIS-AMI provides statistical analysis (also LTI dependent), wherein eye openings are calculated directly from the impulse response, and a time-domain (bit-by-bit) mode, in which results are based on specific input patterns. Figure 14 shows the difference between a time-domain simulation accounting for nonlinear circuit effects, in (a), and a fully linear statistical simulation, in (b). The difference observed in the inner-eye contour suggests that the statistical mode should be applied to the nonlinear DRAM I/O with caution.

Another issue with applying the IBIS-AMI to DDR involves how to handle the forwarded clock in the IBIS-AMI executable model. Traditional IBIS-AMI applications rely on CDR to synchronize the data at the receiver. For DDR, however, the clock is provided by the host as a separate strobe or write-clock (WCK) signal in parallel with the data. One proposed modeling approach is to implement a CDR algorithm to generate a clock within the DDR IBIS-AMI model, but there are concerns over how accurately this approach captures the jitter tracking and unique impairments on the forwarded clock signal. While these challenges will eventually be resolved through new simulation techniques and enhancements to the IBIS-AMI specification, the analysis of DDR5 systems will be much more involved than for previous DDR generations.

LPDDR5

While LPDDR1 was essentially a DDR1 with a lowered supply level, over the generations, LPDDR has incorporated a variety of features and functions, including a suite of new modes of operation, to facilitate generally lower

DDR signaling has evolved tremendously over the last two decades, leading to diversification not only in the architecture of the memory array but also in that of the interface.

power. In parallel with these architectural developments, the LPDDR interface has changed dramatically. As with the other DDR families, LPDDR eventually reached frequencies warranting receive-side termination when attempting to extend LPDDR3 beyond its original target of 1.6-Gb/s/pin, leading initially to a pseudo-open drain I/O solution similar to that adopted by DDR4 and GDDR4/5/5X/6, shown in Figure 15(a) [15]. (Distinct standards are associated with different target voltage levels—1 V, 1.2 V, 1.25 V, and 1.5 V.)

With the added power consumption imposed by termination, LPDDR4 looked for ways to reduce the signaling power with minimal structural change. At the time, it was determined

that transforming the transmitter output stage from the long-standing source-terminated solution to a low-voltage source-terminated logic (LVSTL) architecture, shown in Figure 15(b), could provide significant power reduction in the most critical and common modes of operation (e.g., when the receiver termination is removed and per-pin speed is limited) [16]. Initially, the majority of the power reduction came in the form of the lower ac power ($\propto CV^2f$) associated with charging and discharging the parasitic capacitance of the connection (including the I/O). The natural saturation of the NMOS PU provided a weak clamping mechanism, limiting the swing and thereby reducing one of the Vs in the CV^2f

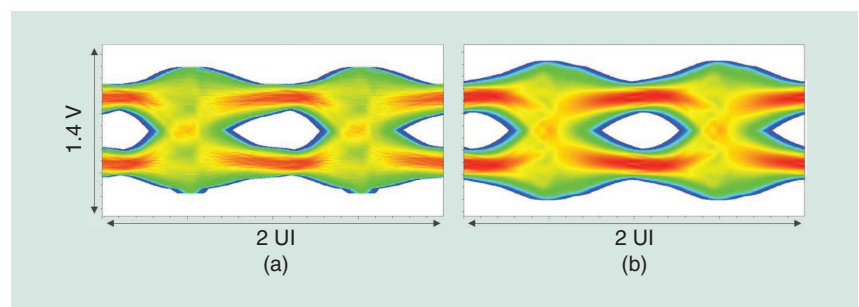


FIGURE 14: A DFE simulation enabled by the IBIS-AMI in (a) bit-by-bit mode and (b) statistical mode.

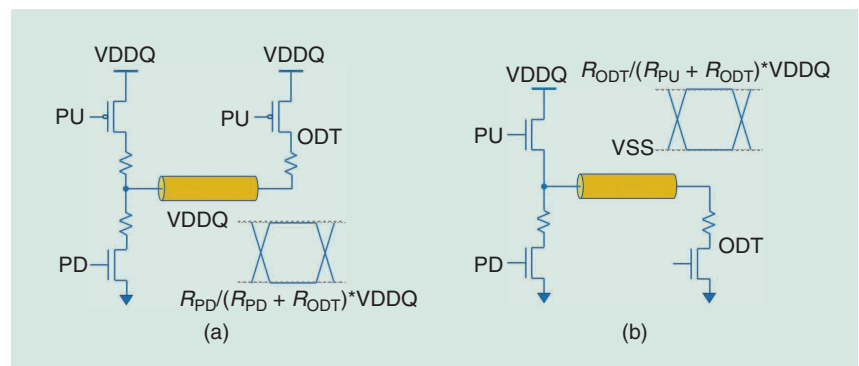


FIGURE 15: The basic signaling topology for (a) DDR4-5/GDDR3-6 (PODL) and (b) LPDDR4-5 (LVSTL) off-chip signaling. ODT: on-die termination.

power. Unfortunately, the weak PU also introduced uncertainty in the high voltage level of the output signal, resulting in a form of ISI. Various circuit techniques helped mitigate this, but the best solution came with the LPDDR4X addendum to the LPDDR4 standard, which required the VDDQ to drop to 0.6 V [17], [18]. This fall in voltage not only reduced the signaling power but simultaneously suppressed much of the PU/PD asymmetry and ISI in the transmitter by significantly limiting the unterminated signal swing.

The LVSTL solution provides even greater flexibility to lower power in that, unlike the pseudo-open drain logic (PODL) architecture employed in

DDR and GDDR, the VSS-terminated interconnect effectively decouples the DRAM I/O voltage supply from that of the controller or host, allowing the controller PHY to optimize its I/O supply level based on its distinct transistor process. This capability not only facilitates power savings but also simplifies reliability within the PHY, where the low-voltage devices of the advanced logic process might otherwise need to be stacked to limit excessive V_{DS} .

To better understand the evolution of the LPDDR output stage and how it came to function best at low voltage-supply levels, Figure 16 plots the inverse-impedance curves for the final N/N output stage of LPDDR4, LPDDR4X, and LPDDR5, with I_{OUT} simulated independently for both the PU and PD mechanisms. As the output node is swept from VSS to VDDQ along the x-axis (LPDDR4 = 1.1 V, LPDDR4X = 0.6 V, and LPDDR5 = 0.5 V), the drive current is reported along the y-axis. The most interesting observation is that the PU current of the LPDDR4 driver (red curve) saturates as the output level exceeds 0.6 V. While this nonlinearity posed little impact on the high-speed, terminated mode of operation because of the near-ground signal levels, when the receiver termination was disabled in low-power mode, a significant degree of circuit-induced ISI resulted from the weak PU straining in the higher voltage range. This result capped the top speed of LPDDR4's low-gear, low-power signaling.

As mentioned, the LPDDR4X addendum reduced the VDDQ to 0.6 V, which simultaneously lowered output power and eliminated the circuit-induced ISI by limiting the signal swing (green curve). Note that the gates of the output stage would continue to be driven between VSS and VDD2 (1.1 V), resulting in more than enough overdrive voltage V_{GS} across the NMOS PU to support a clean low-to-high transition, even at the upper half of the signal range (0.3–0.6 V). LPDDR5 makes an incremental improvement by dropping the output voltage to 0.5 V

and VDD2 to 1.05 V [4], further reducing power while better balancing the PU and PD behaviors for enhanced SI, as shown by the blue curves in Figure 16.

By comparison, Figure 17 presents the inverse impedance of the final stage of the LPDDR5 and DDR5 output drivers. The shaded regions represent the typical operating range established by the dc voltage divider formed between the output impedance (drive strength) and the termination at the far end of the channel (although no termination was included in these simulations). The DDR5 driver exhibits its nonlinear behavior because of the limited voltage headroom at the lower, 1.05-V VDDQ level. In contrast, LPDDR5 exhibits good linearity, despite a VDDQ of 0.5 V [4], which is interesting, given the expected drive limitations of the NMOS PU device unique to the LVSTL architecture.

Thus, as indicated previously with respect to Figure 4, LPDDR5 signaling is not inherently impeded by channel impairments, although applications that increase system memory by connecting multiple DRAMs in parallel tend to degrade SI through the additive channel loading and amplified reflections from the larger effective capacitive input. It is anticipated that LPDDR5 will find itself in such circumstances more often than previous generations as mobile computations increase (e.g., AI-enabled and virtual-reality-supported personal devices) and as nonmobile systems increase the emphasis on power optimization, thus seeking to integrate LPDDR5 rather than the traditional DDR alternative [19]. Channel and loading conditions aside, LPDDR5 is primarily constrained by power, which drives down supply voltage levels (and corresponding voltage margins) while prohibiting compensation through well-established, high-speed, power-hungry circuit techniques. Not only are voltage margins inherently reduced with the new standard, but LPDDR5 is further expected to operate reliably under a degree of dynamic voltage

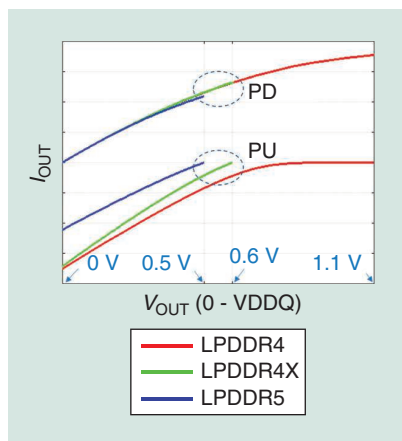


FIGURE 16: Simulated (inverted) output impedance curves for the N/N final driver output stage of the last three iterations of LPDDR (e.g., LPDDR4/4X/5).

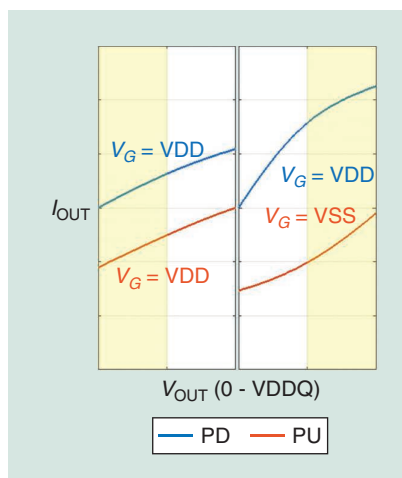


FIGURE 17: The simulated (inverted) output impedance (I/V) of the LPDDR5 and DDR5 output drivers over their respective signal levels.

frequency scaling in both the core and I/O domains [4], which introduces additional complexity.

GDDR6

While the graphics DRAM architecture has traditionally served gaming and high-performance computing, the ongoing explosion of data associated with AI and ML has dramatically broadened the application space for GDDR. Interestingly, as AI and ML find application across an ever-diversifying set of platforms (e.g., mobile devices and home security), so too does GDDR become a viable candidate for use outside the graphics world. Indeed, one of the most compelling growth industries for GDDR is in automotive applications. While it is true that the infotainment and communication segments of automotive computing continue to grow in demand and adoption, less obvious is the extensive computing resources “under the hood” dedicated to semiautonomous driver support systems (e.g., lane recognition and impact avoidance).

Figure 18 highlights the aggressive computing requirements of driver-support systems. While in-vehicle infotainment (IVI) can demand 50- to 300-GB/s of memory bandwidth, advanced driver-assistance systems (ADASs), including object detection and inference, require anywhere from 300-GB/s to 1-TB/s of memory bandwidth, depending on the degree of autonomy. While automotive system requirements have traditionally been addressed with DDR and LPDDR architectures, the demands of these new applications can no longer be so satisfied, simply because of their limited per-chip I/O bandwidth (pin speed $\times$ channel width). As shown, both LPDDR5 or DDR5 could serve most IVI needs, but a fully LPDDR5- or DDR5-populated system would require a prohibitive number of components to support the I/O demands of ADAS functions. GDDR6, with its introductory data rate of 14 Gb/s/pin, stands out as clearly superior in these applications [20], [21].

Reliable signaling at 16 Gb/s across a highly parallel, single-ended channel (32 b or 16 b $\times$ 2 in dual-channel mode) is a nontrivial task requiring exceptional care in circuit, power system, and interconnect design [22]–[25]. Figure 19 provides one simplified example of how circuit and interconnect choices might combine to enable the extreme performance of GDDR6. The figure compares two 16 Gb/s pulse responses corresponding to a typical graphics card channel. In one case, the multilayer printed circuit board (PCB) composing the majority of the interconnect is implemented in a through-hole via technology, introducing channel discontinuities corresponding to the via stubs extending below the

actual routing layer. These discontinuities lead to reflected signal energy, a form of ISI. In the second pulse response, the via stubs are removed through via back-drilling during the manufacturing process or through the use of blind and buried vias. In this case, the pulse peak is higher while the long-term ISI associated with the signal reflections is reduced (both leading to a greater voltage margin). Thus, this manufacturing choice (which comes with a monetary cost) is an effective way to enhance the channel response.

A second method to address the poor quality of the through-hole PCB is to employ active circuits to equalize away the negative characteristics of the channel. This is represented

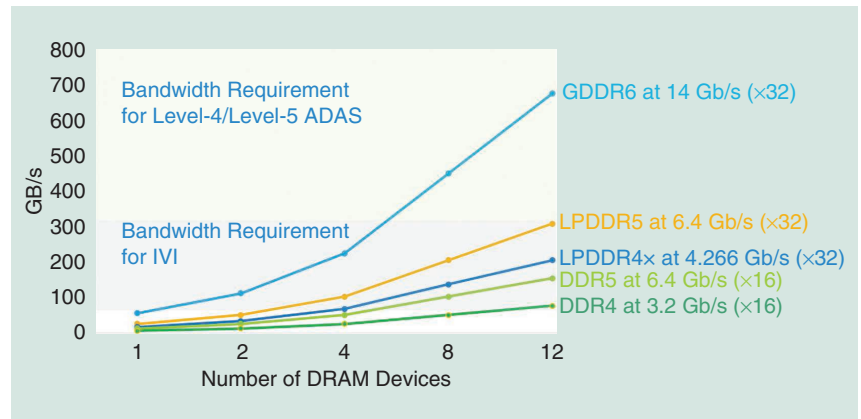


FIGURE 18: The system bandwidth requirements of autonomous and semiautonomous ADASs and IVI.

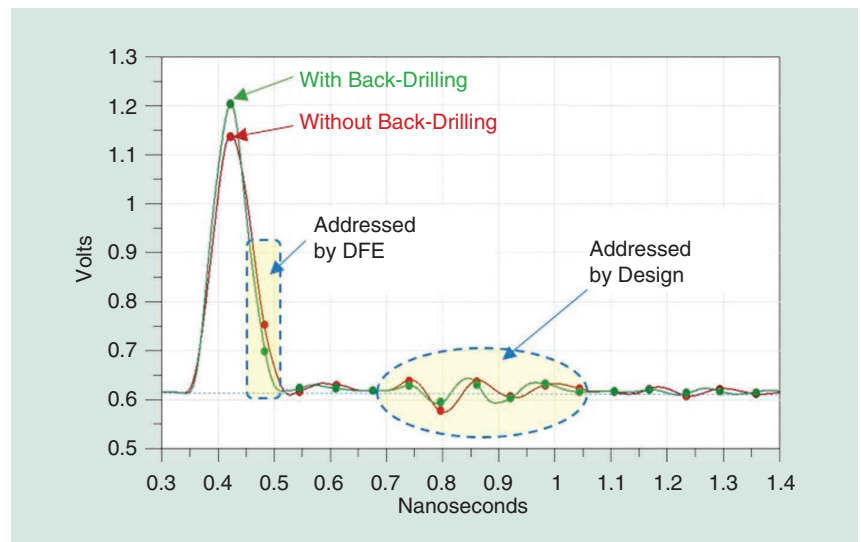


FIGURE 19: A 16-Gb/s pulse response comparison with and without via back-drilling, demonstrating the distinct compensation provided by DFE and back-drilling.

Application-specific channels have influenced the I/O design nearly as much as system power and bandwidth requirements have.

by the highlighted UI over which the single-tap DFE will operate to zero out the near-term ISI. This solution also comes with a cost: added power consumption. Ultimately, it is anticipated that reaching 16 Gb/s with confidence could require both enhancements to the interconnect and the employment of active equalization.

As highlighted in [22] and [23], reaching 16 Gb/s in the DRAM alone (ignoring system influence) is facilitated by an accumulation of design enhancements dating back to the GDDR4 era, including DBI to support power integrity, DFE to address minor channel impairments, a phase-locked loop to filter jitter on the input clock, and regulated voltage

supplies to minimize power-supply-induced jitter along the high-speed WCK path. Each of these played a role in stretching the performance of GDDR5 from its initial target of 5 Gb/s to as high as 9 Gb/s [26], [27]. Additional features further found their way into GDDR5 designs at the higher speeds, including pseudode-emphasis in the off-chip transmitter and frequency-controlled switching of on-DRAM charge pumps to provide greater voltage supply stability [28]. Careful conditioning of the WCK included input equalization (e.g., CTLE based), duty cycle correction, and more prolific use of current-mode-logic-based design techniques to further minimize sensitivity to supply

fluctuations throughout the internal clock network [23].

The GDDR5 and GDDR5X standards did something unique when specifying the output drive strength and input termination values of the I/O. Rather than leave the drive and on-die termination (ODT) flexible, as in the DDR and LPDDR standards, which allow for driver and ODT adjustment through enabling/disabling combinations of calibrated 240- Ω PU and PD legs (seven legs for DDR5 and six legs for LPDDR5), the recent graphics standards fixed the drive strength and ODT settings [29], [30]. By removing flexibility while still supporting full termination disablement, the layout of the GDDR circuitry was simplified, leading to reduced bandwidth-limiting parasitics. Furthermore, the graphics standards chose to specify an imbalanced PU/PD impedance ratio to maximize the fundamental signal swing. As mentioned previously and shown in Figure 15, the swing of the transmitted signal in the GDDR topology is a function of the driver PD and the ODT. Thus, the PU/PD drive-strength ratio of 60 Ω /40 Ω , coupled with the 60- Ω ODT to VDDQ at the far end of the channel, increased the available signal window by 135 mV over a 50- Ω /50- Ω solution.

While this unbalanced drive/termination combination offered advantages, it also introduced challenges. One particular challenge was in the successful implementation of transmitter equalization. As described in [28], a pseudode-emphasis function was implemented (see Figure 20) to compensate for near-term ISI in the channel. As shown in the schematic, the de-emphasized behavior, rather than occurring on a UI boundary in keeping with the traditional implementation, was triggered by a gate delay τ , tuned to be less than one UI at the highest targeted data rate. To form a symmetric output, in the context of the unbalanced PU/PD drive strength and termination, the de-emphasis strength must also be skewed, as indicated in the figure.

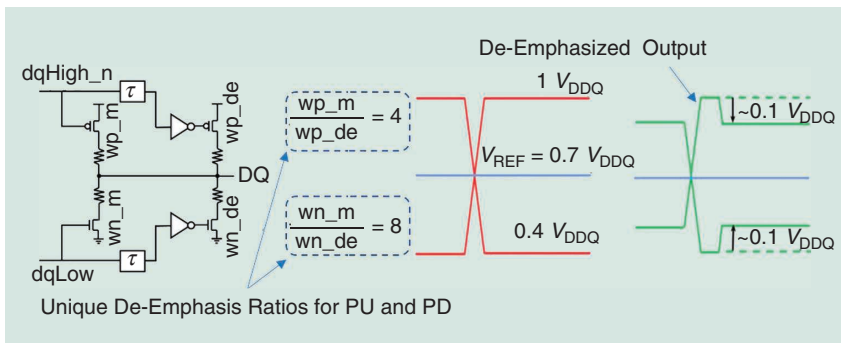


FIGURE 20: A simplified schematic of the pseudode-emphasis scheme used to partially equalize the high-speed data readout from the DRAM, with wp_m and wn_m representing the primary (or main) driver legs and wp_de and wn_de representing the additional leg(s), which provides the de-emphasis functionality.

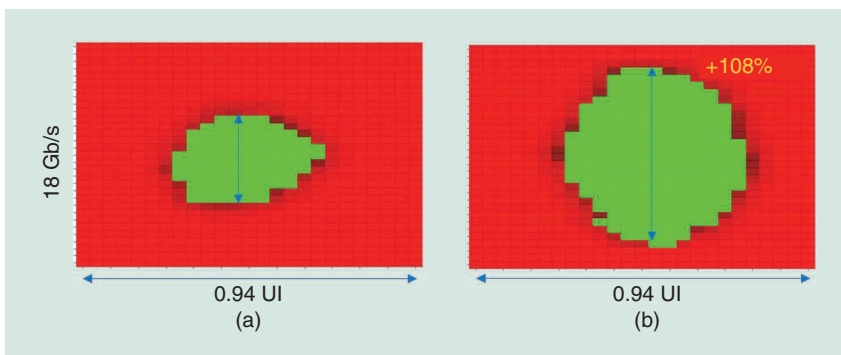


FIGURE 21: Measured GDDR6 DRAM output data eye shmoos showing the impact of de-emphasis: (a) without de-emphasis and (b) with de-emphasis.

While this approach is much simpler than the traditional embodiment and imposes a lesser parasitic capacitance penalty, it produces a temporary offset (mismatch) in reverse termination, as the drive strength dynamically adjusts from a stronger to a weaker value to form the de-emphasized waveforms shown. A second potential limitation of this approach is the process, voltage, and temperature dependence of the gate delay, suggesting that the duration of the emphasis might change in situ unless compensated for. But, as is shown in the measured 18-Gb/s GDDR6 data eye shmoos captured in a GDDR6 test and development platform and reported in Figure 21, this compact de-emphasis scheme is very effective.

While GDDR6 grandfathered in the 60- Ω /40- Ω PU/PD and termination solution, it was determined that, at the higher speeds of the new standard, a better match to the channel impedance would be helpful. Hence, GDDR6 supports both 60- and 48- Ω PU and ODT strength, with 40- Ω PD. Adding this flexibility has the potential to increase the unfavorable parasitics at the sensitive high-speed output node of the driver. However, it has been observed (and is demonstrated in Figure 22) that the added benefit of the stronger termination far outweighs the relatively minor capacitance tradeoff as the signaling speed pushes higher. In addition to reducing reflected ISI and increasing the bandwidth of the channel, 48- Ω termination provides the added benefit of raising the signal common-mode level, thereby increasing V_{GS} at the NMOS inputs of the receiver, which is very helpful in the DRAM process at these speeds. That said, the added flexibility imposes complexity on both the receiver and transmitter, in that the GDDR6 receiver must accommodate multiple termination values, and the transmitter must support the additional drive combination in both the main output circuitry and in the de-emphasis ratios discussed with reference to Figure 20.

As suggested previously, successful system-level integration of GDDR6

The broader DDR family looks to continue to push the boundaries of single-ended signaling into the future.

requires much more than a working DRAM. Complementary enhancements in the PCB routing strategy and corresponding robust modeling and simulation are essential as well. Rules of thumb sufficient for earlier generations cannot be applied without careful scrutiny, including, for example, full 3D channel modeling to characterize each channel boundary crossing (e.g., from planar routes on the PCB to the vertical via field to the package balls and vias and finally to the package routing and die

termination), as shown in Figure 23. Careful channel study over a variety of board configurations (e.g., thickness, stack-up, and materials) indicates that, once the ISI is minimized through impedance matching from segment to segment, the vertical transitions in the break-out regions below the components become the largest channel contributors to data eye degradation.

While the package ball grid is fixed by the JEDEC definition, there is some flexibility in the via positioning below

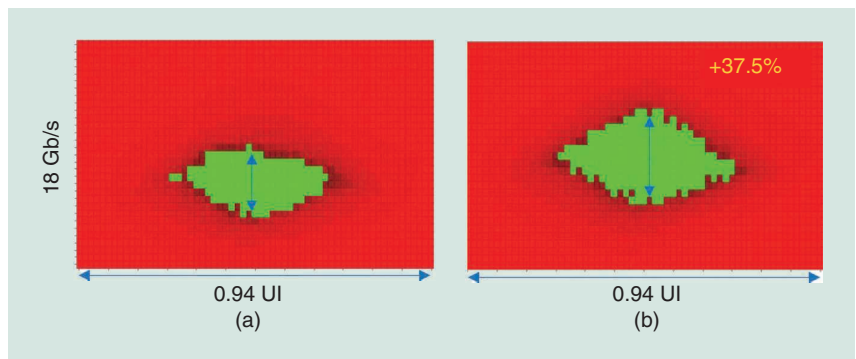


FIGURE 22: Measured GDDR6 DRAM input data eye shmoos showing the impact of stronger termination, unavailable in previous GDDR generations: (a) 60- Ω termination and (b) 48- Ω termination.

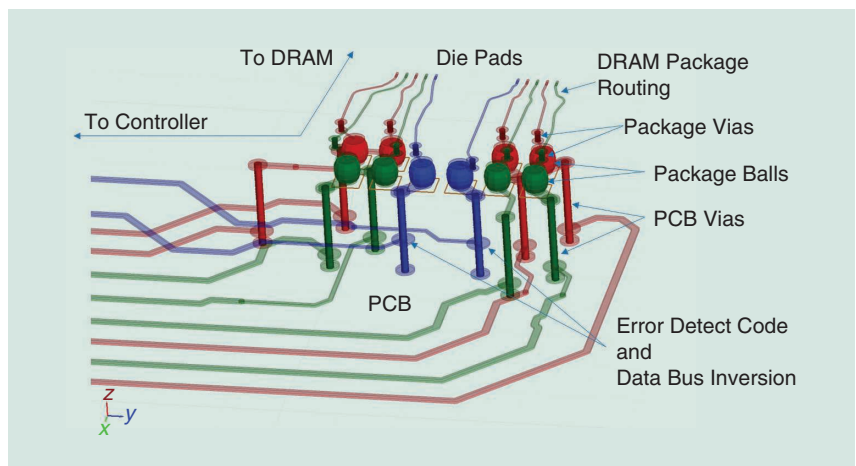


FIGURE 23: A 3D modeling example of the GDDR5X-to-PCB interconnect. Such detailed modeling is critical to support high-speed, single-ended signaling because the bulk of the channel impairment occurs at the vertical interfaces from chip to board at both ends of the channel.

the component, as shown in the figure. Using the traditional dog-bone layout approach, it is possible to separate the majority of the vertical transitions while introducing power and ground signal return vias between the signal vias for dramatically improved SI. Figure 24 compares the field strength/coupling between balls in the GDDR6 ball grid [Figure 15(a) and (b)] with a fictitious checkerboard solution [Figure 15(c)] and reports a significant coupling reduction, translating into a 15.8-dB decrease in crosstalk at the worst-case ball location. This experiment suggests that coupling between vias (which are significantly taller and therefore more sensitive than the package balls) may be mitigated through thoughtful via

layout, including generous introduction of signal return vias. As the PCB vias below the DRAM were identified as one of the most sensitive knobs in the channel performance, it is understandable that steps to minimize the via impact are an effective component of GDDR6 enablement (high-speed single-ended). As suggested in Figure 18, back-drilling of the tall vias in the main PCB is another key to increasing signaling margins. Removal of the via stubs impacts both ISI (by eliminating a major source of signal reflections) and crosstalk (by doing away with the crosstalk associated with the stub-reflected signaling energy). The simulated 16-Gb/s data eyes presented in Figure 25 provide a compelling

case for via back-drilling and, at the same time, highlight the potentially additive benefits of combining back-drilling with the GDDR6 one-tap DFE. The additive nature of these distinct forms of compensation can be understood by observing that they operate (to a degree) on different aspects of the channel pulse response, as observed in Figure 19. In addition, via back-drilling addresses crosstalk, which the DFE does not explicitly do. As speed increases, so does the first postcursor of the channel pulse response, thus increasing the effectiveness of the one-tap DFE. At the same time, increases in speed drive more signal energy across the resonance formed by the through-hole

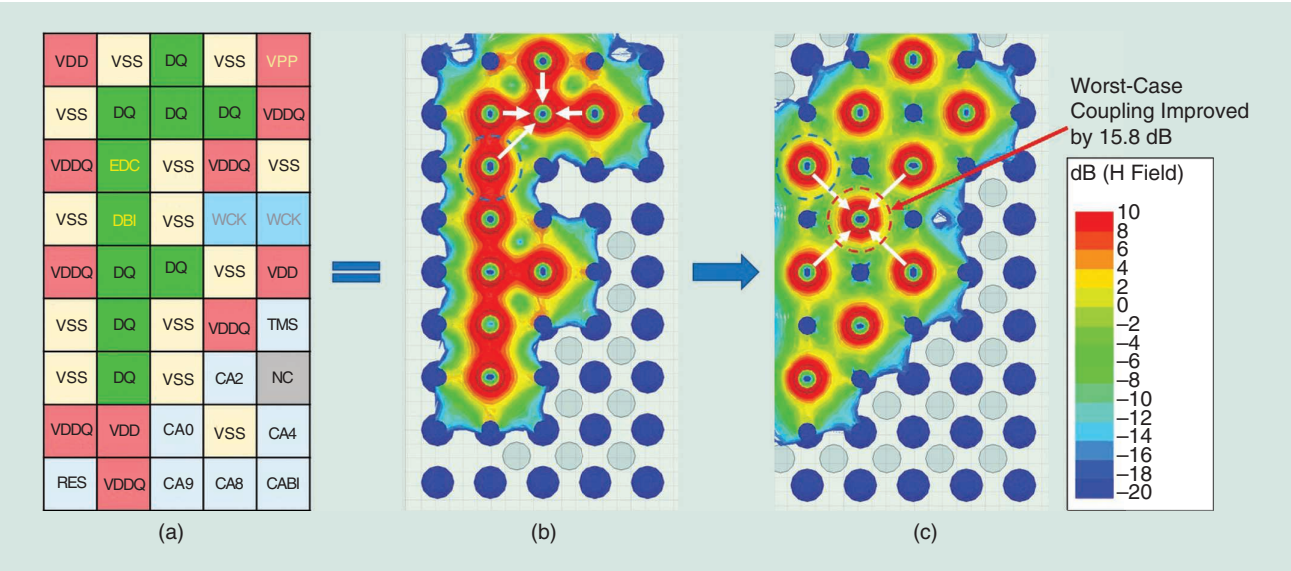


FIGURE 24: A coupling simulation example beginning from (a) the JEDEC-defined GDDR6 package ball-out (b) to the simulated fields between GDDR6 package balls (c) to the fields between balls for a fictitious checkerboard solution.

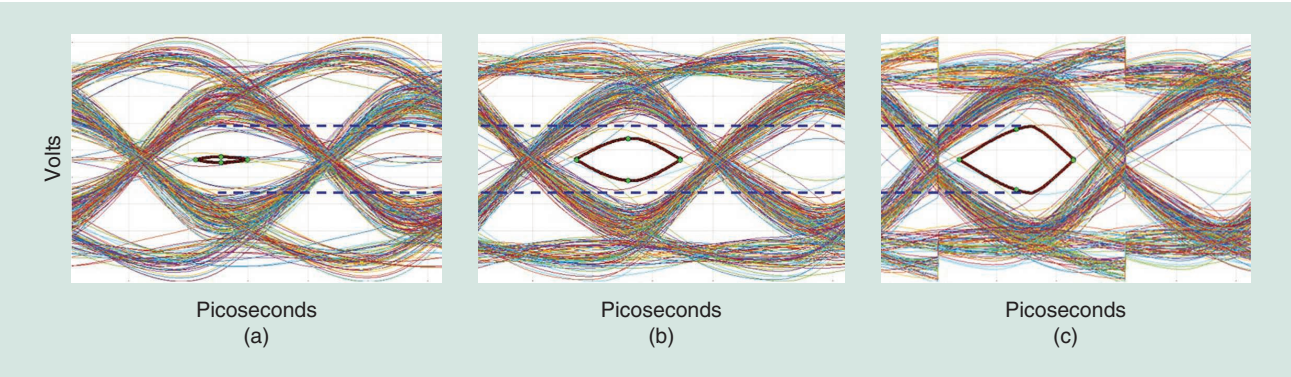


FIGURE 25: Peak-distortion-based data eyes corresponding to the worst-case data lines in an exemplary GDDR6 system channel: (a) baseline (no back-drilling, no DFE), (b) back-drilled (no DFE), and (c) back-drilled plus one-tap DFE.

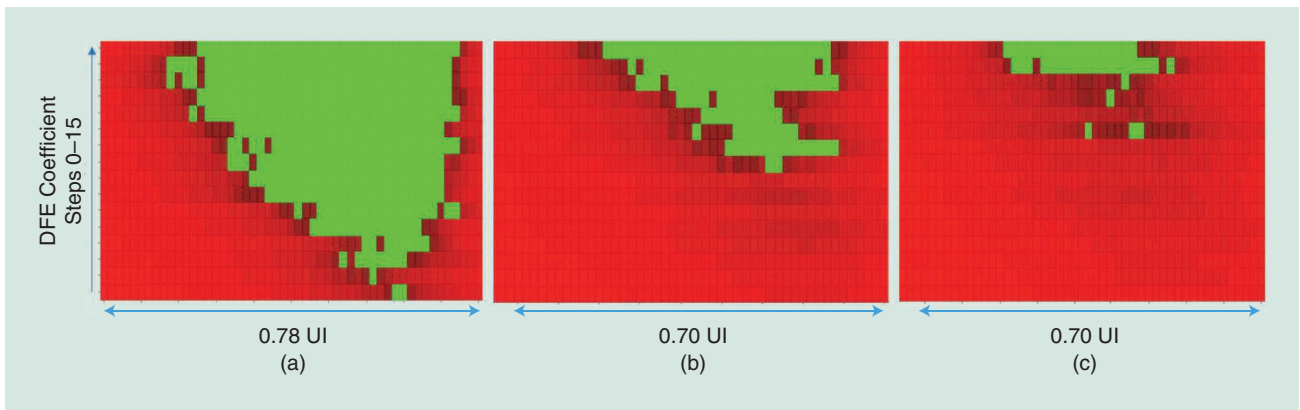


FIGURE 26: Measured GDDR6 DRAM input data eye shmoos at (a) 18 Gb/s, (b) 19 Gb/s, and (c) 20 Gb/s, showing the need to scale the DFE coefficient strength with speed.

via stub, leading to increases in both long-term ISI and crosstalk, effectively mitigated through back-drilling. The pass/fail results presented in Figure 26 are consistent with a first postcursor scaling with speed. As shown, stepping the data rate from 18 Gb/s to 19 Gb/s to 20 Gb/s requires a larger DFE coefficient to achieve a passing margin across the interface.

Conclusion

As stated in the introductory section, DDR signaling has evolved tremendously over the last two decades, leading to diversification not only in the architecture of the memory array but also in that of the interface. Application-specific channels have influenced the I/O design nearly as much as system power and bandwidth requirements have. The growing impact of the multidrop server channel, along with a broad range of target environments, has led the DDR branch to incorporate multitap DFE, while shrinking supply voltages to facilitate low-power operation led LPDDR to completely rethink the output driver structure. At the same time, GDDR has reached speeds requiring nearly equal care of the external channel and the chip itself. With all these adaptations and many others not captured here, the broader DDR family looks to continue to push the boundaries of single-ended signaling into the future. For additional details, see the perspective on the DRAM interface in [31].

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